Biomedical Image Processing

SEGMENTATION – PART I

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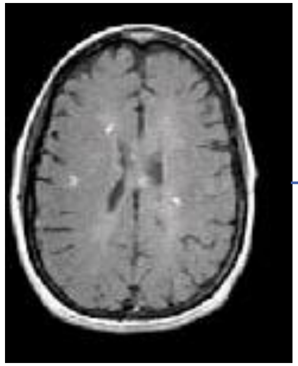
Medical Image Processing

PIPELINES FOR DELINEATION OR DETECTION OF ANATOMICAL STRUCTURES IN MEDICAL IMAGING

Transformed image

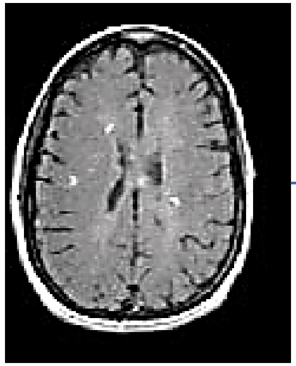
Binary image with objects

Original image → **Preprocessing** → **Segmentation method** → **Post-processing** → **Segmented image**



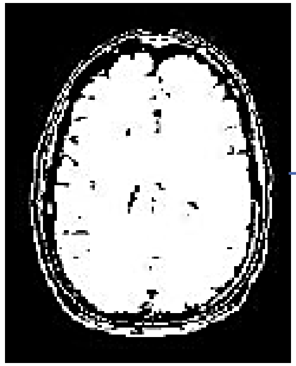
Improve the image (eliminate noise, enhance details)

- Local operations
- Global Operations
- Filtering
- Mathematical morphology



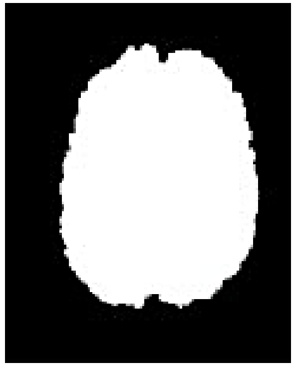
Find the object of interest (binary image)

- Threshold
- Deformable models
- Classification models
- Deep learning, ...



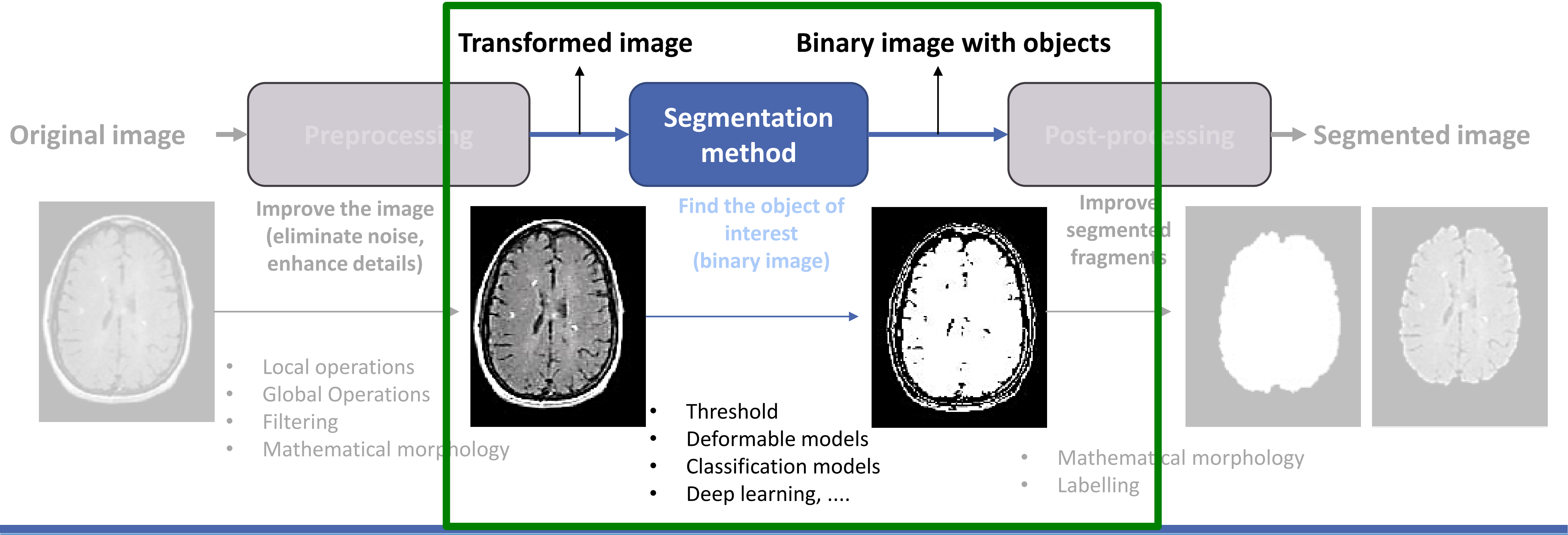
Improve segmented fragments

- Mathematical morphology
- Labelling



Medical Image Processing

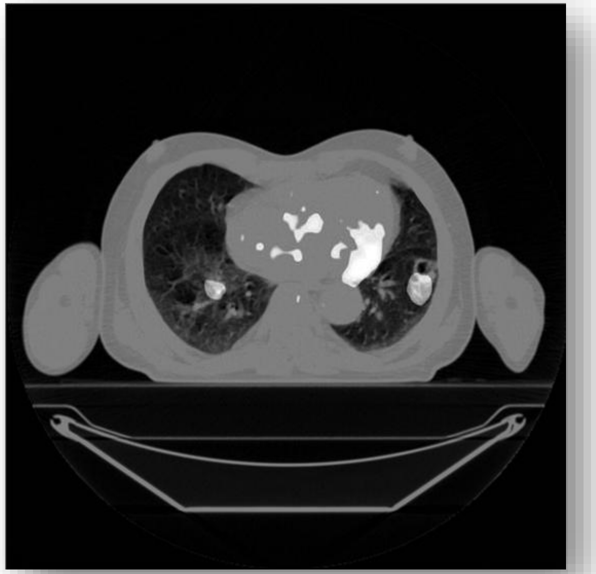
PIPELINES FOR DELINEATION OR DETECTION OF ANATOMICAL STRUCTURES IN MEDICAL IMAGING

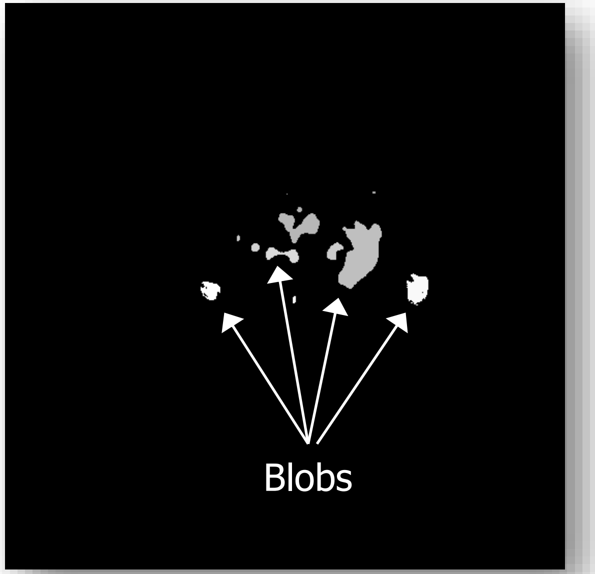


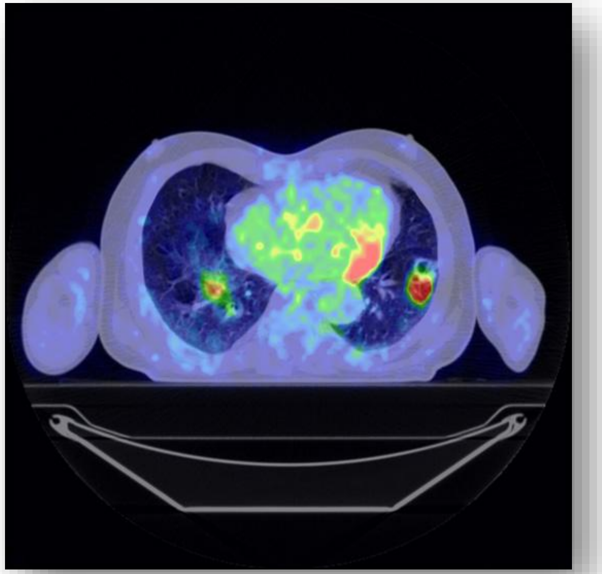
Segmentation: concepts

BLOBS

- In image segmentation it is common to use the term blob to define a region of an image whose pixels have a certain common characteristic
- They can be defined by analyzing the similarity of the intensity of the brightness or its color.



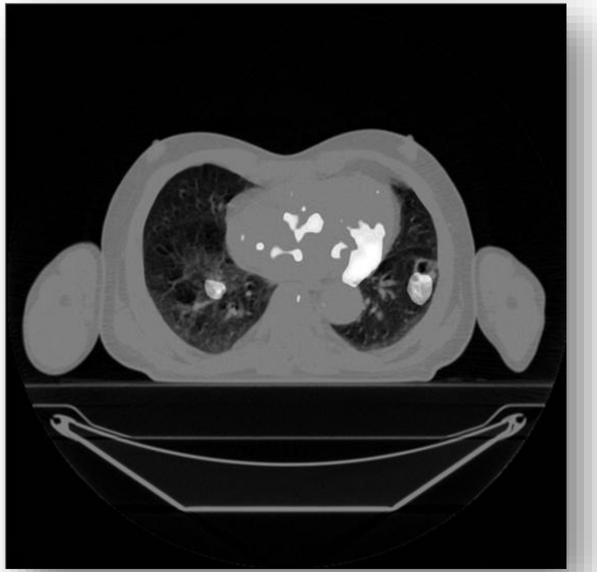




Segmentation: concepts

- ❑ Delineate an object for later extraction of attributes and calculation of descriptive parameters
- ❑ The segmentation process results in two types of pixels:
  - Foreground, which represents the segmented object
  - Background
- ❑ The result is a binary image
- ❑ In medical imaging, segmentation is used for the recognition of anatomical structures
- ❑ The selection of an appropriate targeting technique depends on the type of images and the application

Segmentation: concepts

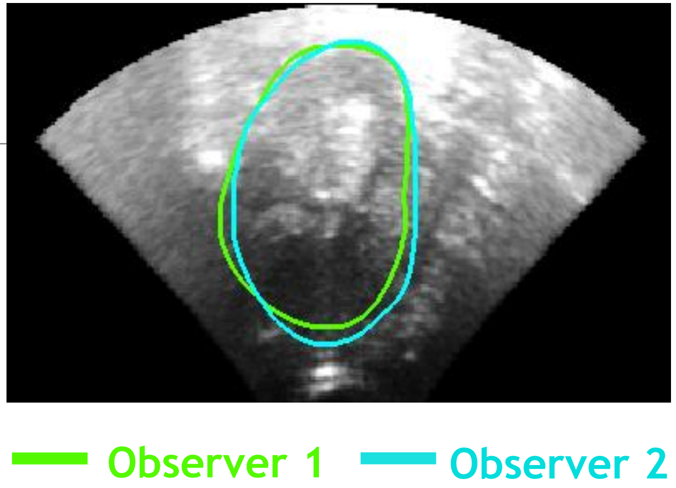


Segmentation: concepts

EXAMPLES OF MEDICAL APPLICATIONS

- Creation of anatomical models
- Simulation of surgeries
- Measurement of volume of tumors and organs
- Automatic classification of blood cells
- Detection of microcalcifications
- Diagnosis
- Intraoperative use
- …

Segmentation: concepts



CLASSIFICATION OF SEGMENTATION TECHNIQUES

- Manual segmentation
  - A user elaborates the delineation manually
  - In 3D it is necessary the 2D delineation of the structure slice to slice
- Semi-automatic segmentation
  - Requires user interaction to start the targeting process
  - Click to define structure origin, bounding-box drawing around the structure, …
- Automatic segmentation
  - Does not require intervention from a user
  - Segmentation is completely automatic

Segmentation

SEGMENTATION METHODS

- Threshold
- Region growing
- Watershed
- Active contours
- Deformable models
- Registration/Atlas
- Learning methods (Next lecture)
  - Unsupervised
    - *Clustering, k-means, …*
  - Supervised
    - *Support Vector Machines* (SVM)
    - *Random Forests*
    - *Machine learning*
      - Neural networks
      - *Deep learning*

Segmentation

THRESHOLDING

- Used in images where the object to be segmented has different intensity from the other elements of the image
- Segmentation method:

$$\begin{cases} B(x,y) = 1, if\ I(x,y) > th \\ B(x,y) = 0, if\ I(x,y)\ <= th \end{cases}$$

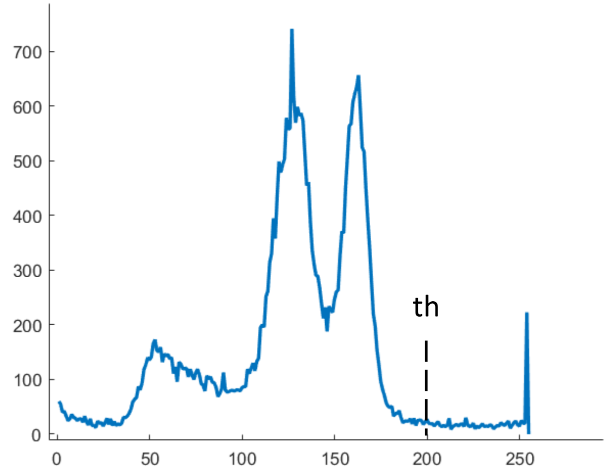
- The threshold th value can be set manually by the user or calculated automatically
- Since it results in a binary image, it can be called binarization

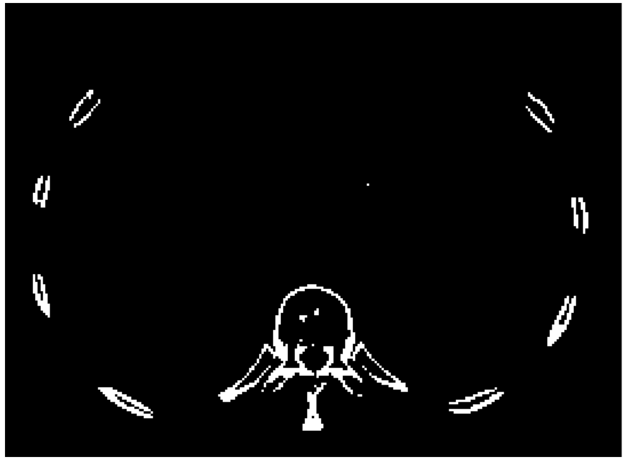
Segmentation

THRESHOLDING

- Example – Segmentation of bone

```
ImgBin = np.where(Img<th,0,1);
```





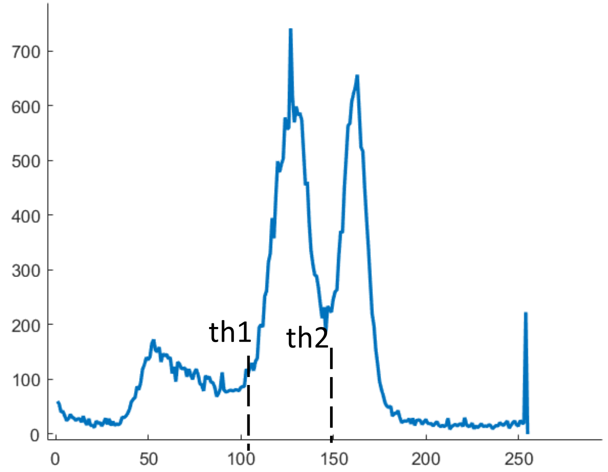
Note: For visualization, the values corresponding to 0 were not presented in the histogram

Segmentation

THRESHOLDING

- Example – Segmentation of lesions

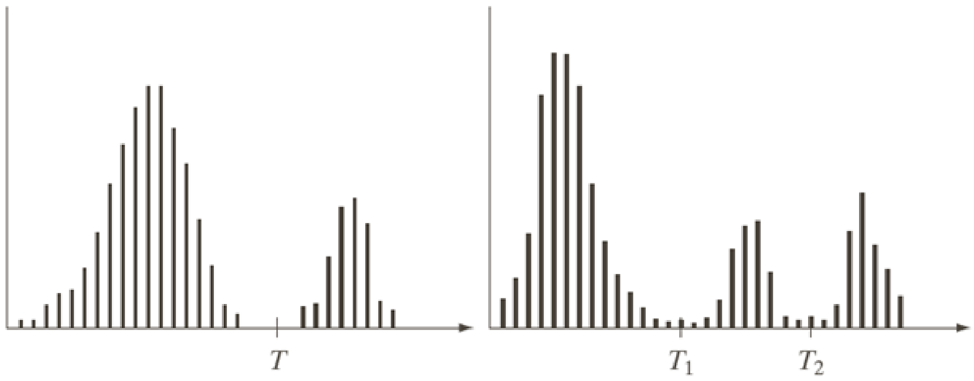
```
ImgBin = np.where(np.logical_and(Img>th1, Img<th2), 1, 0);
```



Segmentation

THRESHOLDING SEGMENTATION USING THE *OTSU* METHOD

- Threshold value set automatically
- Attempts to separate the image into different regions by minimizing variance in each region
- Uses the histogram to find distributions for each region
- Can separate the image into more than two regions



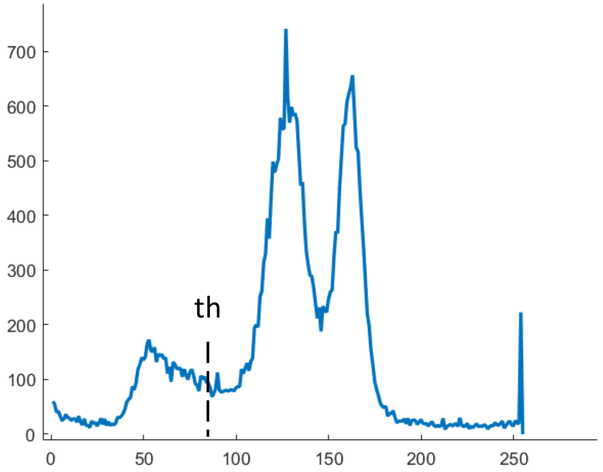
Segmentation

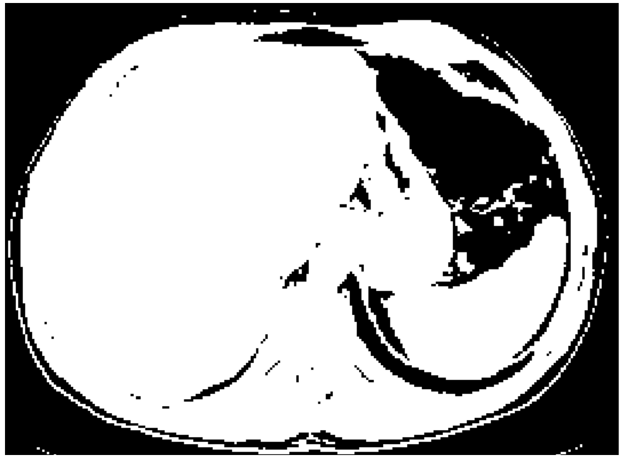
THRESHOLDING SEGMENTATION USING THE *OTSU* METHOD

- Example

```
from skimage.filters import threshold_multiotsu

thresholds = threshold_multiotsu(image,1)
regions = np.digitize(image, bins=thresholds)
```





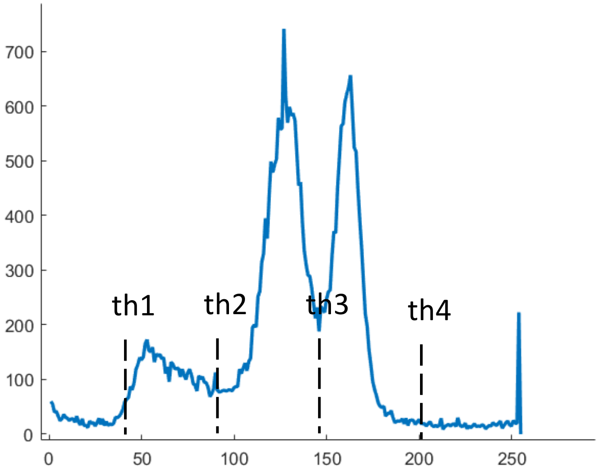
Segmentation

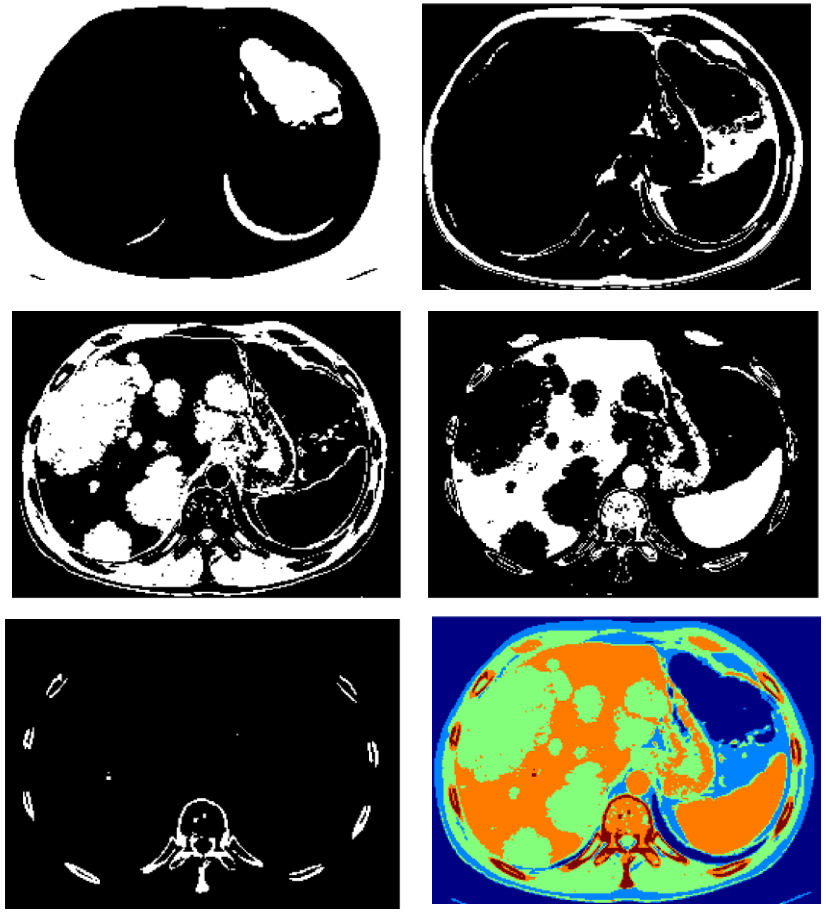
THRESHOLDING SEGMENTATION USING THE *OTSU* METHOD

- Example

```
from skimage.filters import threshold_multiotsu

thresholds = threshold_multiotsu(image,5)
regions = np.digitize(image, bins=thresholds)
```





Segmentation

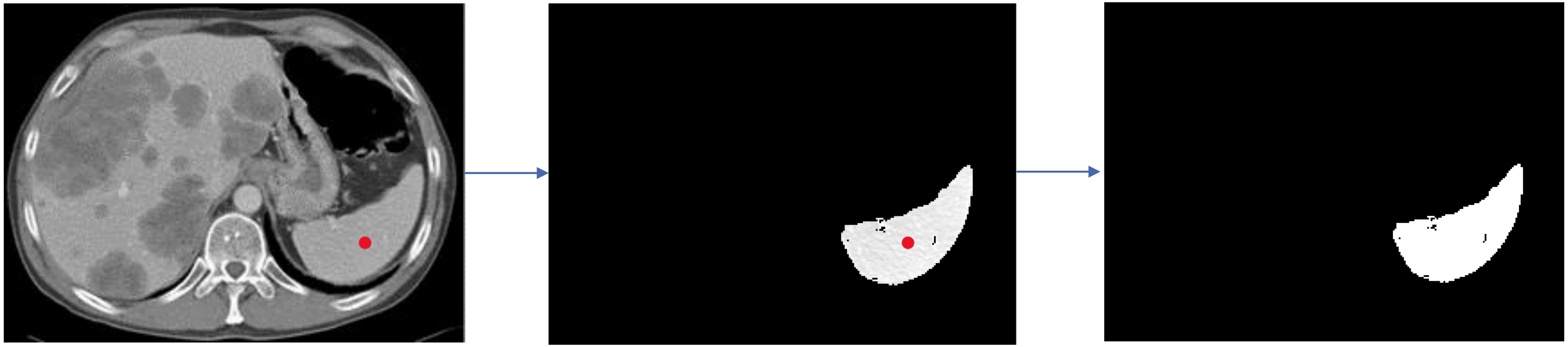
REGION GROWING

- Groups regions according to a certain criteria
- Pixel aggregation begins with a set of points (seed) marked in one or a set of regions
- The region begins to grow by associating each seed with pixels in its vicinity that have similar properties, such as same level of gray, texture, color, shape.,…

Segmentation

REGION GROWING

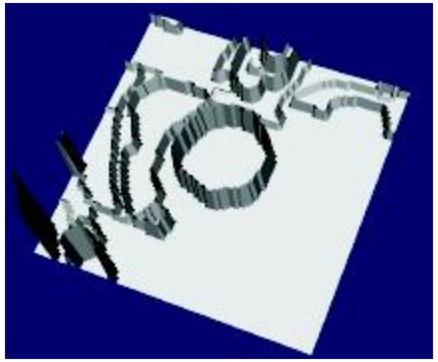
- Example – Segmentation of the spleen

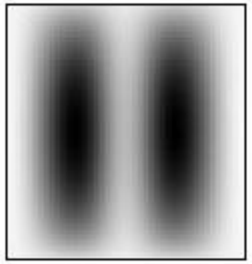


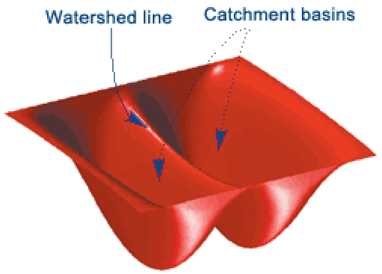
Segmentation

WATERSHED

- *Watershed*
  - Crest that divides areas drained by different river systems
  - Consider the image as a topographic surface
  - Lighter values correspond to peaks
  - Darker values are valleys
  -
- Application to image processing
  - Assume that the image is a surface
  - Simulate progressive surface flooding
  - Regions that drain to different sites belong to different objects
  - Different regions of the image correspond to different levels



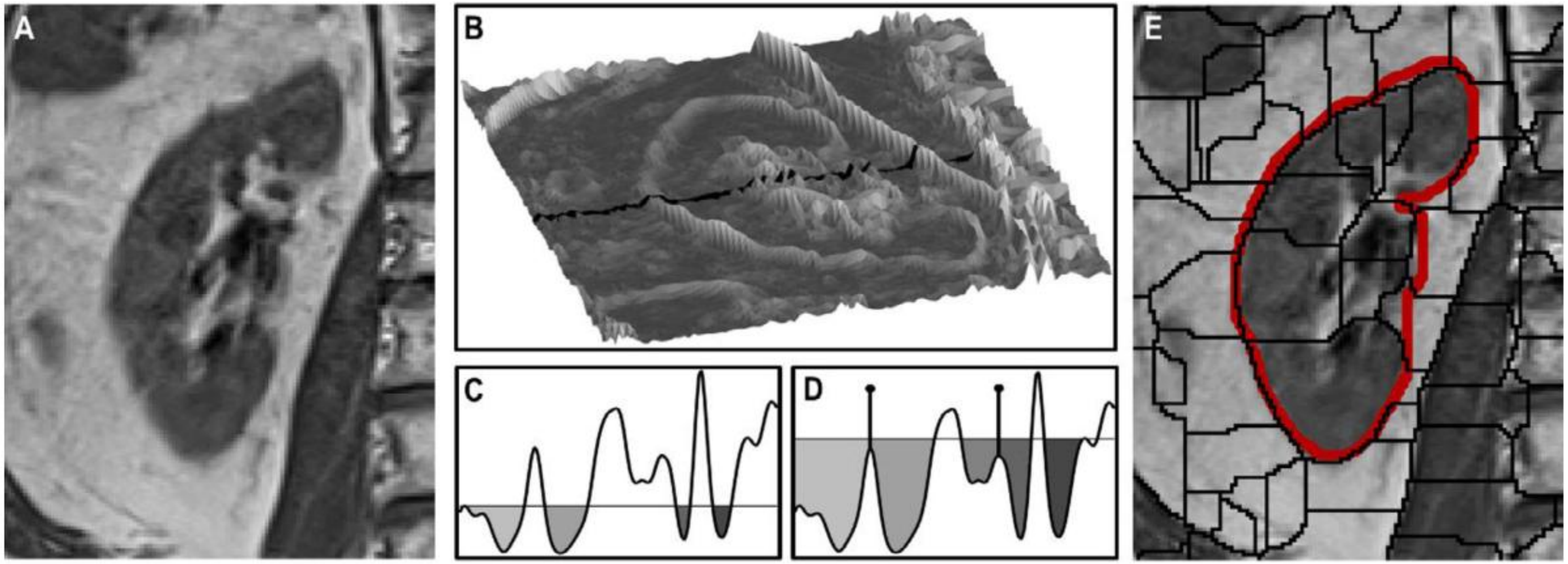




Segmentation

WATERSHED

- ❑ Example



Segmentation

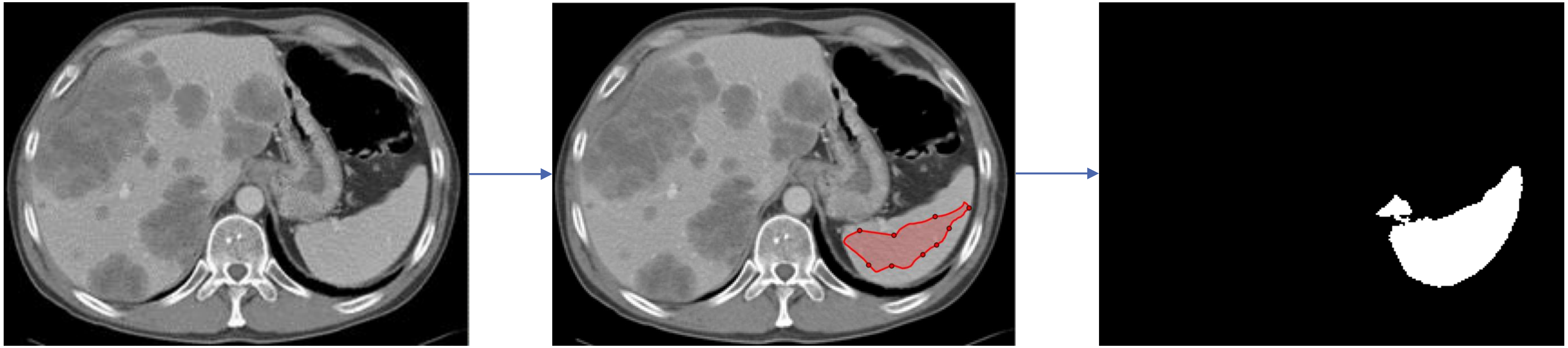
ACTIVE CONTOURS (SNAKES)

- Active contours
  - Evolution from an initial contour to the edge of an object by minimizing an energy
- It is necessary to create an initial outline
  - Manually
  - Using an automatic method
- Often used as a method of refinement
- Common energy types:
  - edge-based: the contour is evolved until it finds contours
  - based on intensity (chan-vese): the contour is evolved until the image is divided into two homologous regions, not allowing the object to be segmented to have a variation in intensities.

Segmentation

ACTIVE CONTOURS (SNAKES)

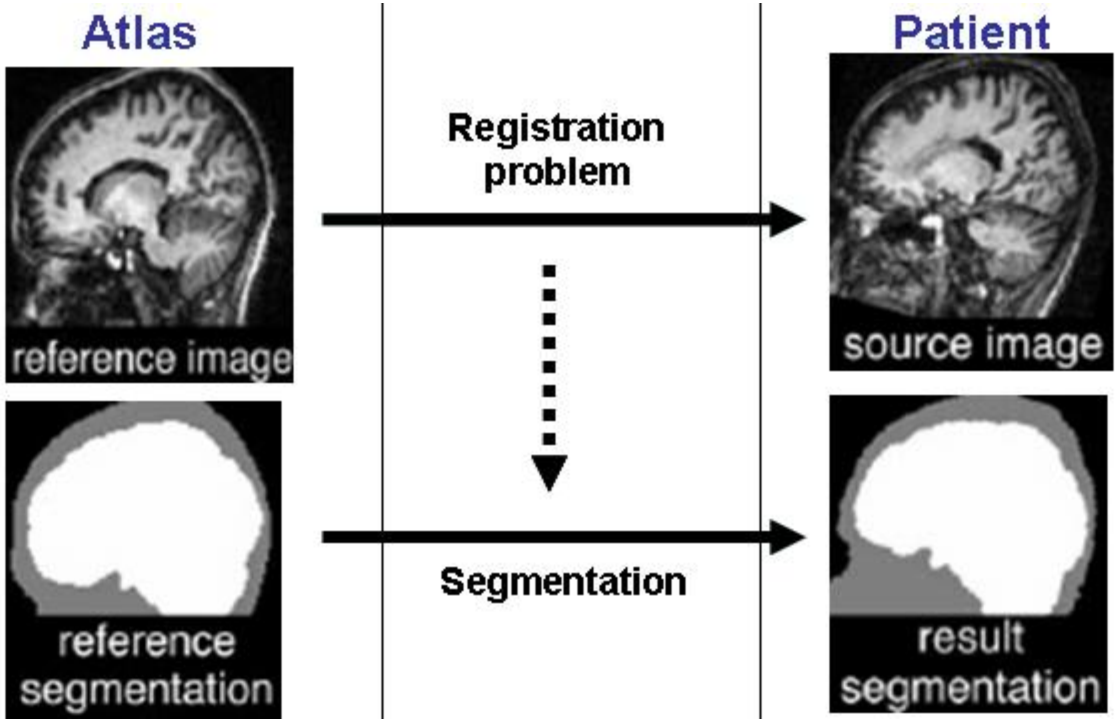
- Example - Segmentation of the spleen

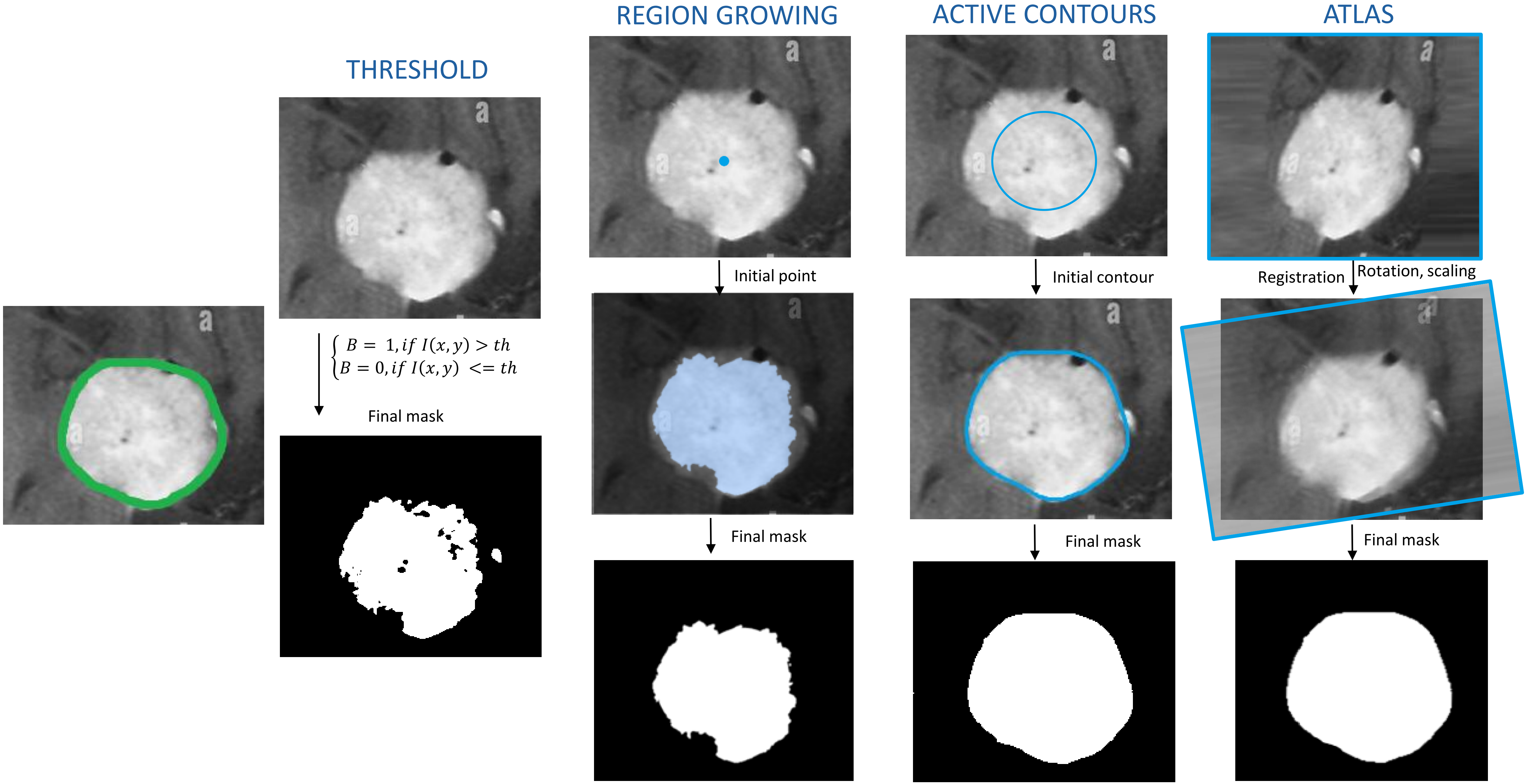


Segmentation

ATLAS/REGISTRATION

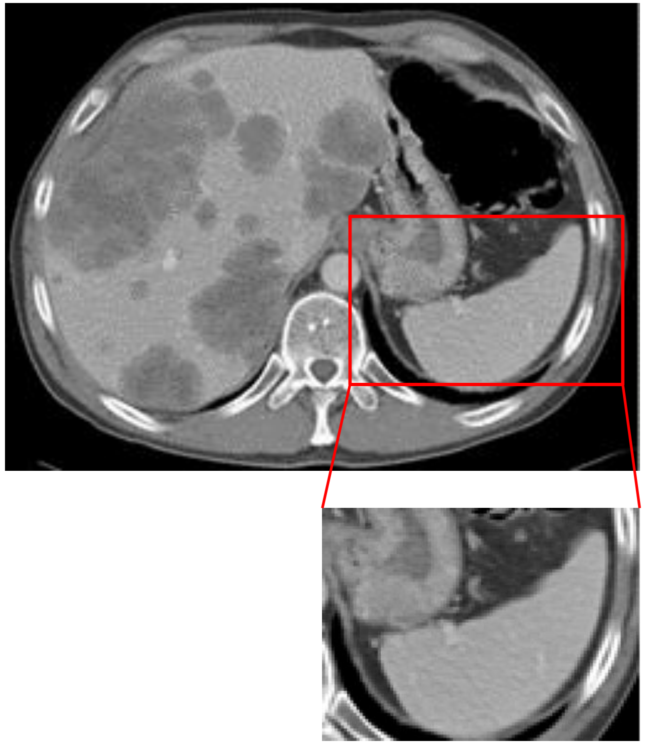
- ❑ Involves aligning a labelled patient's scan (atlas) to a new image through image registration
- ❑ After the alignment, a transformation is found and applied to the reference segmentation
- ❑ The anatomical structures segmentation are transferred to the new image, segmenting the new image.




THRESHOLD
REGION GROWING
ACTIVE CONTOURS
ATLAS
$\begin{cases} B = 1, if\ I(x,y) > th \\ B = 0, if\ I(x,y) <= th \end{cases}$
Final mask
Initial point
Final mask
Initial contour
Final mask
Registration
Rotation, scaling
Final mask

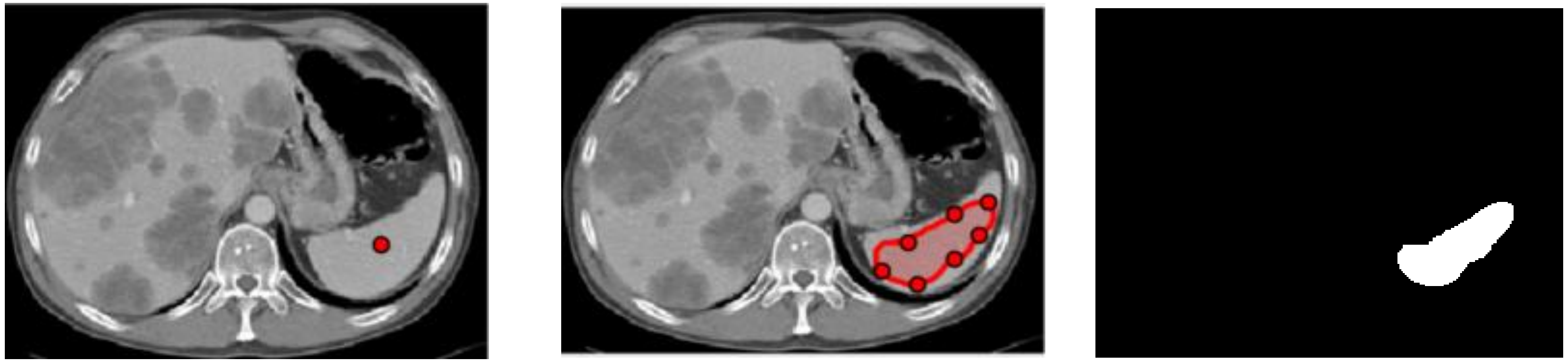
User interaction

- Ideally, segmentation methods should be automatic to avoid variability between users;
- However, there are situations where user interaction is necessary or improves segmentation.;
- Restriction of the region of interest
  - Creating a box to shrink the image to be analyzed



User interaction

- Ideally, segmentation methods should be automatic to avoid variability between users.;
- However, there are situations where user interaction is necessary or improves segmentation.;
- Initialization of a method
  - Give a starting point or an initial outline to start a method (can also be effected during the segmentation pipeline)
  - drawpoint, drawfreehand, drawcircle



Semi(automatic) Segmentation

- Automatic or semi-automatic?
  - Threshold with *th* defined manually for each image
  - Threshold with *th* pre-defined manually (afterward the value is maintained)
  - Active contour where the initial contour is defined manually
  - Active contour where the initial contour is retrieved from a threshold method
  - Atlas where the user defines a bounding box around the organ to segment
  - Region growing where the initial point is set automatically
  - Region growing where the inicial point is defined by an user

Segmentation evaluation

❑EVALUATION OF A SEGMENTATION METHOD

- ❑ The performance of a pipeline to segment an object is evaluated using evaluation metrics;
- ❑ These metrics are calculated by comparing the result of the segmentation with a known ground-truth;
- ❑ In medical imaging, ground-truth should be performed by experienced physicians or radiologists;

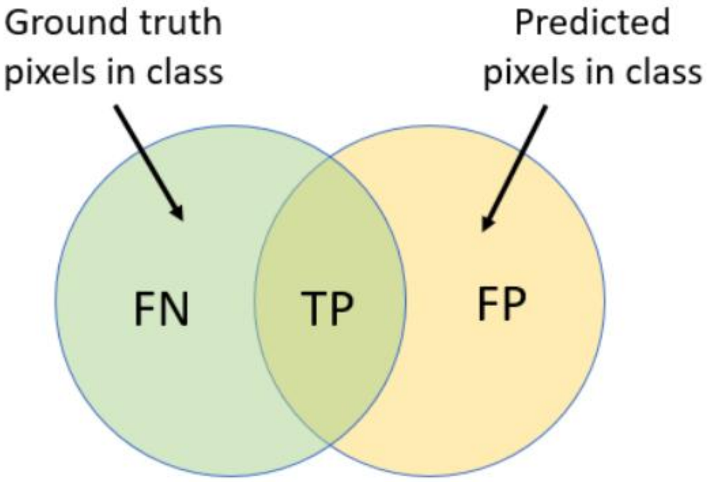
Ground-truth- performed manually by an expert

Segmentation result

Segmentation evaluation

EVALUATION OF A SEGMENTATION METHOD

- TP – True Positives;
- TN – True Negatives;
- FP – Falses Positives;
- FN – Falses Negatives;



Dice

$$DSC = \frac{2TP}{2TP + FP + FN}$$

Accuracy

$$\mathrm{ACC} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{P} + \mathrm{N}} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}}$$

Recall

$$\mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{P}} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}$$

Precision

$$\mathrm{PPV} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}$$

IoU

$$\mathrm{TS} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN} + \mathrm{FP}}$$

		Actual	
		Positive	Negative
Predicted	Positive	True Positive	False Positive
	Negative	False Negative	True Negative

Segmentation evaluation

❑EVALUATION OF A SEGMENTATION METHOD

❑ ***Dice***

- The Dice coefficient measures the overlap between the predicted and ground truth masks, measuring their similarity with a value between 0 (no overlap) and 1 (perfect match)

❑ ***Intersection-over-union***

- IoU calculates the ratio of the intersection to the union of predicted and ground truth areas, assessing the overall agreement of the segmented regions

❑ ***Accuracy***

- Accuracy represents the proportion of correctly classified pixels (both positive and negative) out of the total number of pixels in the image

❑ ***Precision***

- Precision measures the proportion of true positive pixels among all pixels predicted as positive, indicating the reliability of positive predictions

❑ ***Recall***

  ❑ Recall quantifies the proportion of true positive pixels that were correctly identified, reflecting the method's ability to detect all relevant areas